CS 341: Foundations of Computer Science II
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Homework 11

1. Answer each part TRUE or FALSE.
   (a) $2n = O(n)$.
   (b) $n^2 = O(n)$.
   (c) $n^2 = O(n \log^2 n)$.
   (d) $n \log n = O(n^2)$.
   (e) $3^n = O(2^n)$.
   (f) $3^n = 2^{O(n)}$.
   (g) $2^{2^n} = O(2^{2^n})$.

2. Let $b > 1$ be a constant. Show that $O(t(n)) \times O(b^{t(n)}) = 2^{O(t(n))}$.

3. (a) Show that P is closed under union.
   (b) Show that P is closed under concatenation.
   (c) Show that P is closed under complementation.

4. A **triangle** in an undirected graph is a 3-clique. Show that $TRIANGLE \in$ P, where $TRIANGLE = \{ \langle G \rangle \mid G \text{ contains a triangle} \}$.

5. Using the polynomial-time algorithm for context-free language recognition (i.e., the CYK algorithm or dynamic programming), fill out the table for string $w = abba$ and CFG G:

$$\begin{aligned} S &\rightarrow RT \\ R &\rightarrow TR \mid a \\ T &\rightarrow TR \mid b \end{aligned}$$